

O2 — Organic Chemistry II

1. Aldehydes and Ketones

1.1 Structure and Properties

Both aldehydes and ketones contain the **carbonyl group (C=O)**. In aldehydes, the carbonyl carbon is bonded to at least one hydrogen atom (it is always at the end of the chain). In ketones, the carbonyl carbon is bonded to two carbon atoms (it is within the chain).

The C=O bond is strongly polar ($\delta^+C=O\delta^-$) because oxygen is much more electronegative than carbon (O = 3.5, C = 2.5). This makes the carbonyl carbon electron-deficient and susceptible to attack by nucleophiles. The polarity of C=O also means aldehydes and ketones have higher boiling points than alkanes of similar mass (dipole-dipole forces), but lower than alcohols (cannot form hydrogen bonds with themselves, though they can accept hydrogen bonds from water).

Physical properties:

- Methanal (HCHO) is a gas at room temperature; ethanal (CH₃CHO) is a volatile liquid (bp 20°C).
- Short-chain aldehydes and ketones are soluble in water because C=O can hydrogen-bond with water.
- Solubility decreases as the hydrocarbon chain lengthens.

1.2 Nucleophilic Addition

The characteristic reaction of aldehydes and ketones is **nucleophilic addition**. The nucleophile attacks the δ^+ carbonyl carbon, and the π bond in C=O breaks, with both electrons moving to the oxygen.

Reaction with NaBH₄ (sodium borohydride) — Reduction

NaBH₄ provides the hydride ion (H⁻) as the nucleophile. This is a reduction reaction (addition of hydrogen/removal of oxygen):



Conditions: NaBH_4 in aqueous solution or methanol at room temperature. NaBH_4 is a mild, selective reducing agent (it reduces $\text{C}=\text{O}$ in aldehydes and ketones but does not reduce $\text{C}=\text{C}$ or carboxylic acids; LiAlH_4 also reduces carboxylic acids).

Mechanism: The H^- ion (from BH_4^-) attacks the δ^+ carbon of the $\text{C}=\text{O}$ group. The π bond breaks and both electrons move to oxygen, forming an alkoxide intermediate ($\text{R}-\text{CH}_2-\text{O}^-$). The alkoxide is then protonated by water to give the alcohol product.

Reaction with HCN (hydrogen cyanide) — Nucleophilic Addition



Conditions: HCN with a trace of NaCN or KCN as a catalyst. The CN^- ion is the actual nucleophile (HCN itself is too weak a nucleophile). The reaction must be carried out in a fume cupboard because HCN is extremely toxic.

Mechanism (two steps):

- Nucleophilic addition:** The lone pair on CN^- attacks the δ^+ carbon of $\text{C}=\text{O}$. A curly arrow shows the lone pair on CN^- forming a new $\text{C}-\text{CN}$ bond. Simultaneously, the $\text{C}=\text{O}$ π bond breaks and both electrons move to the oxygen, creating an alkoxide intermediate ($\text{C}-\text{O}^-$).
- Protonation:** The alkoxide ion (O^-) is protonated by HCN to form the $-\text{OH}$ group, regenerating CN^- (which acts catalytically).

Importance of this reaction: It extends the carbon chain by one carbon. The nitrile group ($-\text{CN}$) can be subsequently:

- Hydrolysed** to a carboxylic acid: $\text{RCN} + 2\text{H}_2\text{O} + \text{H}^+ \rightarrow \text{RCOOH} + \text{NH}_4^+$ (dilute acid, reflux)
- Reduced** to a primary amine: $\text{RCN} + 4[\text{H}] \rightarrow \text{RCH}_2\text{NH}_2$ (LiAlH_4 in dry ether, or H_2/Ni catalyst)

Chirality: If the carbonyl compound is an aldehyde (or an unsymmetrical ketone), the hydroxynitrile product contains a new chiral centre. A **racemic mixture** is produced because the planar carbonyl group can be attacked equally from either side by the nucleophile.

Exam Tip: In nucleophilic addition mechanism questions, you must show: (1) a curly arrow from the lone pair on CN^- to the δ^+ carbonyl carbon, (2) a curly arrow from the $\text{C}=\text{O}$ π bond to the oxygen, (3) the intermediate alkoxide ion with O^- clearly shown, (4) protonation by HCN in a separate step. Show the δ^+ and δ^- charges on the $\text{C}=\text{O}$ bond. State that a racemic mixture forms and explain why (attack from either side of the planar $\text{C}=\text{O}$).

1.3 Distinguishing Aldehydes from Ketones

Aldehydes can be oxidised to carboxylic acids (they have a hydrogen on the carbonyl carbon that can be lost). Ketones cannot be further oxidised under normal conditions. This difference is the basis for distinguishing them:

Test	Reagent	Result with aldehyde	Result with ketone
Tollens' test	$[\text{Ag}(\text{NH}_3)_2]^+$ (ammoniacal silver nitrate), warm in water bath	Silver mirror forms on the glass ($\text{Ag}^+ \rightarrow \text{Ag}$)	No reaction (stays colourless)
Fehling's test	Cu^{2+} in NaOH /tartrate solution, heat	Blue solution $\rightarrow$ brick-red precipitate of Cu_2O	No reaction (stays blue)
Acidified $\text{K}_2\text{Cr}_2\text{O}_7$	$\text{K}_2\text{Cr}_2\text{O}_7 / \text{H}_2\text{SO}_4$, warm	Orange $\rightarrow$ green (aldehyde oxidised to carboxylic acid)	No reaction (stays orange)

2,4-DNPH test (Brady's reagent): Both aldehydes AND ketones react with 2,4-dinitrophenylhydrazine to give an orange/yellow precipitate (a 2,4-dinitrophenylhydrazone). This test confirms the presence of a $\text{C}=\text{O}$ group but does not distinguish between aldehydes and ketones. The precipitate can be recrystallised and its melting point compared with known values to identify the specific compound.

2. Carboxylic Acids and Derivatives

2.1 Carboxylic Acids

Carboxylic acids contain the -COOH (carboxyl) functional group. They are **weak acids** — they partially dissociate in water to establish an equilibrium:



The carboxylate ion ($\text{RCOO}^{\text{-}}$) is stabilised by delocalisation of the negative charge over both oxygen atoms, which drives the equilibrium to the right to some extent. However, the equilibrium still lies well to the left — carboxylic acids are only partially ionised (typical K_{a} values are $\sim 10^{-5} \text{ mol dm}^{-3}$).

Reactions of carboxylic acids (typical acid reactions):

- **With reactive metals:** $2\text{CH}_3\text{COOH} + \text{Mg} \rightarrow (\text{CH}_3\text{COO})_2\text{Mg} + \text{H}_2$ (effervescence)
- **With carbonates:** $2\text{CH}_3\text{COOH} + \text{Na}_2\text{CO}_3 \rightarrow 2\text{CH}_3\text{COONa} + \text{H}_2\text{O} + \text{CO}_2$ (effervescence — this is a test for carboxylic acids)
- **With bases (neutralisation):** $\text{CH}_3\text{COOH} + \text{NaOH} \rightarrow \text{CH}_3\text{COONa} + \text{H}_2\text{O}$
- **With alcohols (esterification):** See section 2.2

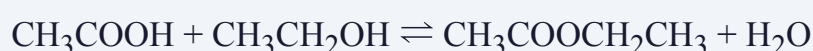
Reducing carboxylic acids: Carboxylic acids are difficult to reduce. LiAlH_4 (a strong reducing agent in dry ether) is needed to reduce them to primary alcohols: $\text{RCOOH} + 4[\text{H}] \rightarrow \text{RCH}_2\text{OH} + \text{H}_2\text{O}$. NaBH_4 is too weak.

2.2 Esters

Esters contain the -COO- linkage and are formed from the reaction of a carboxylic acid with an alcohol (or from acyl chlorides/acid anhydrides with alcohols).

Naming esters: The alcohol part comes first (as an alkyl group), followed by the acid part with the suffix -oate. Example: ethyl ethanoate ($\text{CH}_3\text{COOCH}_2\text{CH}_3$) is from ethanol + ethanoic acid.

Formation from carboxylic acid + alcohol:



Conditions: concentrated H_2SO_4 catalyst, heat under reflux. This is a reversible reaction with a low yield unless excess of one reactant is used.

Hydrolysis of esters (breaking the ester bond):

Type	Conditions	Products	Reversible?
Acid hydrolysis	Dilute H_2SO_4 or HCl , reflux	Carboxylic acid + alcohol	Yes (reversible equilibrium)
Base (alkaline) hydrolysis	Dilute NaOH , reflux	Sodium salt of carboxylic acid + alcohol	No (irreversible, goes to completion)

Base hydrolysis is irreversible because the carboxylate ion (RCOO^-) formed is very stable (resonance-stabilised) and does not react back with the alcohol. This reaction is also called **saponification** when applied to fats and oils (triglycerides) to make soap (sodium salts of fatty acids).

2.3 Acyl Chlorides

Acyl chlorides (R-COCl) are very reactive carboxylic acid derivatives. The C=O carbon is highly δ^+ due to the combined electron-withdrawing effects of both the O and Cl atoms. They react vigorously with nucleophiles at room temperature via a **nucleophilic addition-elimination** mechanism.

Nucleophile	Reaction	Products	Observations
Water (H_2O)	$\text{RCOCl} + \text{H}_2\text{O} \rightarrow \text{RCOOH} + \text{HCl}$	Carboxylic acid	White/steamy fumes of HCl
Alcohol (R'OH)	$\text{RCOCl} + \text{R'OH} \rightarrow \text{RCOOR'} + \text{HCl}$	Ester	White fumes of HCl
Ammonia (NH_3)	$\text{RCOCl} + 2\text{NH}_3 \rightarrow \text{RCONH}_2 + \text{NH}_4\text{Cl}$	Primary amide	White fumes; excess NH_3 needed
Primary amine (R'NH_2)	$\text{RCOCl} + 2\text{R'NH}_2 \rightarrow \text{RCONHR'} + \text{R'NH}_3^+\text{Cl}^-$	N-substituted amide	Second equivalent of amine acts as base

Why acyl chlorides are preferred over carboxylic acids for ester/amide synthesis:

- Reactions are **faster** (more reactive C=O due to Cl electron withdrawal)

- Reactions go to **completion** (irreversible — HCl leaves as gas)
- Higher yields (>95% compared to ~67% equilibrium yield for acid + alcohol)
- No catalyst needed

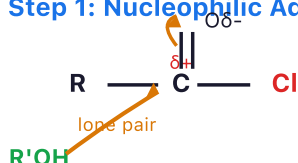
Addition-elimination mechanism (e.g. with alcohol):

1. **Addition:** The lone pair on the oxygen of R'OH attacks the δ^+ carbonyl carbon. The C=O π bond breaks, electrons move to oxygen, forming a tetrahedral intermediate.
2. **Elimination:** The C-O⁻ reforms the C=O bond, and the Cl⁻ is expelled as the leaving group.
3. The H on the oxygen then transfers (or is lost as HCl with the Cl⁻).

Acyl Chloride: Addition-Elimination Mechanism



Step 1: Nucleophilic Addition



Tetrahedral intermediate:



Step 2: Elimination From tetrahedral intermediate: O⁻ lone pair reforms C=O
Cl⁻ expelled as leaving group RCOOR' + HCl (ester + hydrogen chloride)
Addition of nucleophile → tetrahedral intermediate → elimination of Cl⁻

2.4 Acid Anhydrides

Acid anhydrides (RCO-O-COR) undergo similar reactions to acyl chlorides but are less vigorous, cheaper, and safer to use. They are preferred in industry for these reasons:





The by-product is a carboxylic acid rather than HCl, which is less corrosive and easier to handle. Acid anhydrides are also less moisture-sensitive than acyl chlorides.

Exam Tip: Acyl chlorides react with all four nucleophiles (water, alcohol, ammonia, amine). Acid anhydrides react with the same nucleophiles but more gently. Know that acyl chlorides produce HCl as a by-product (white steamy fumes) while anhydrides produce a carboxylic acid. In synthesis questions, if asked to make an ester or amide efficiently, choose an acyl chloride or acid anhydride rather than a carboxylic acid.

3. Aromatic Chemistry

3.1 Structure of Benzene

Benzene (C_6H_6): A planar, cyclic molecule with a ring of 6 carbon atoms. Each carbon is sp^2 hybridised, forming three σ bonds (two to adjacent carbons, one to hydrogen) in a trigonal planar arrangement. The remaining unhybridised p orbital on each carbon overlaps sideways with its neighbours, creating a continuous ring of delocalised π electron density above and below the plane.

Evidence against the Kekule model (alternating single and double bonds):

- **Bond lengths:** All C-C bonds in benzene are the same length (0.140 nm), intermediate between a C-C single bond (0.154 nm) and a C=C double bond (0.134 nm). Kekule's model predicts alternating long and short bonds.
- **Enthalpy of hydrogenation:** The expected value for "cyclohexatriene" with three isolated C=C bonds would be $3 \times -120 = -360 \text{ kJ mol}^{-1}$. The actual value is only -208 kJ mol^{-1} , which is 152 kJ mol^{-1} less exothermic. This 152 kJ mol^{-1} represents the **delocalisation (stabilisation/resonance) energy** — benzene is thermodynamically more stable than predicted.
- **Reaction behaviour:** Benzene does not readily undergo addition reactions (which would destroy the delocalised system). Instead, it preferentially undergoes **electrophilic substitution**, which preserves

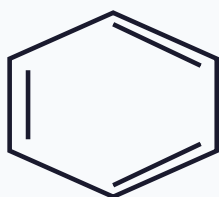
the aromatic ring and its stability.

- **Electron density maps:** X-ray diffraction and electron density maps show uniform electron density around the ring, not alternating regions.

Benzene Structure: Kekule vs Delocalised Model

Kekule Model

(Incorrect)



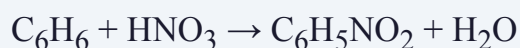
- ✗ Predicts 2 bond lengths
- ✗ Predicts addition reactions
- ✗ ΔH

ΔH_{hyd} should be -360 kJ/mol vs Delocalised Model (Correct) ✓ All bonds equal (0.140 nm) ✓ Substitution, not addition ✓
 $\Delta H_{\text{hyd}} = -208$ kJ/mol (152 kJ more stable) 6 p-orbitals overlap → delocalised π system → extra thermodynamic stability

3.2 Electrophilic Substitution Reactions

Benzene's delocalised π system is a region of high electron density that attracts electrophiles. However, to preserve the thermodynamic stability of the aromatic ring, benzene undergoes **substitution** (not addition).

Nitration



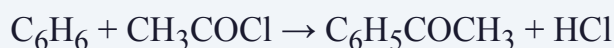
Conditions: Concentrated HNO_3 + concentrated H_2SO_4 (as catalyst), heated to 50°C . Temperature must be kept below 55°C to prevent further substitution (di- and tri-nitrobenzene).

Electrophile: The nitronium ion NO_2^+ , generated by the protonation of HNO_3 by H_2SO_4 :



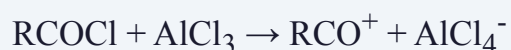
Mechanism (3 steps):

1. The π electrons of the benzene ring attack the electrophilic NO_2^+ . A curly arrow shows the π electrons forming a new bond to nitrogen.
2. A positively charged, non-aromatic intermediate forms (the Wheland or arenium intermediate). The ring has lost its aromaticity — one carbon is now sp^3 with both H and NO_2 attached.
3. A proton (H^+) is lost from the carbon bearing the NO_2 group. The electron pair from the C-H bond restores the aromatic π system. The H^+ reacts with HSO_4^- to regenerate H_2SO_4 (catalyst).

Friedel-Crafts Acylation

Conditions: Acyl chloride with AlCl_3 catalyst (halogen carrier/Lewis acid), reflux under anhydrous conditions.

Electrophile: The acylium ion RCO^+ , generated when AlCl_3 accepts a lone pair from the Cl of the acyl chloride:



The mechanism is identical to nitration: the π electrons attack RCO^+ , an intermediate forms, then H^+ is lost to restore aromaticity. AlCl_3 is regenerated: $\text{AlCl}_4^- + \text{H}^+ \rightarrow \text{AlCl}_3 + \text{HCl}$.

Friedel-Crafts Alkylation

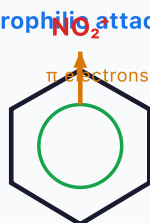
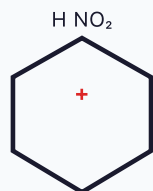
Conditions: Halogenoalkane with AlCl_3 catalyst, reflux. The electrophile is R^+ (carbocation).

Limitation: Alkylation can lead to polysubstitution because the alkyl group donates electrons to the ring (activating it towards further electrophilic attack). Acylation is preferred because the $\text{C}=\text{O}$ group deactivates the ring, preventing further substitution and giving monosubstituted products.

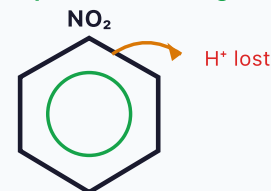
Electrophilic Aromatic Substitution (Nitration)

Step 1:

 Generate electrophile: $\text{HNO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{NO}_2^+ + \text{HSO}_4^- + \text{H}_2\text{O}$
 NO_2^+ = nitronium ion (the electrophile)

Step 2: Electrophilic attack

Wheland Intermediate


Ring lost aromaticity (not stable)

Step 3: Restore ring


Aromaticity restored

Overall: $\text{C}_6\text{H}_6 + \text{HNO}_3 \rightarrow \text{C}_6\text{H}_5\text{NO}_2 + \text{H}_2\text{O}$ (conc. $\text{HNO}_3/\text{H}_2\text{SO}_4$, 50°C)

 H_2SO_4 acts as catalyst: regenerated when H^+ reacts with HSO_4^-

 Friedel-Crafts uses same mechanism: AlCl_3 generates R^+ or RCO^+ electrophile

3.3 Phenol

Phenol ($\text{C}_6\text{H}_5\text{OH}$) is benzene with a hydroxyl group directly attached to the ring. The lone pair on the oxygen is partially delocalised into the ring, increasing the electron density of the π system.

Consequences of delocalisation:

- Phenol is **more reactive towards electrophilic substitution** than benzene. The increased electron density activates the ring, so reactions occur under milder conditions.
- Phenol is **more acidic than aliphatic alcohols** ($\text{pK}_a = 10$ vs ~ 16 for ethanol). The phenoxide ion ($\text{C}_6\text{H}_5\text{O}^-$) is stabilised by delocalisation of the negative charge into the ring. However, phenol is still a weak acid (much weaker than carboxylic acids, $\text{pK}_a \sim 5$).
- Phenol does NOT react with Na_2CO_3 (unlike carboxylic acids) but does react with NaOH : $\text{C}_6\text{H}_5\text{OH} + \text{NaOH} \rightarrow \text{C}_6\text{H}_5\text{O}^-\text{Na}^+ + \text{H}_2\text{O}$

Bromination of phenol (comparison with benzene):



Phenol reacts with **bromine water at room temperature** (no catalyst needed) to give a white precipitate of 2,4,6-tribromophenol. Compare with benzene, which requires a halogen carrier catalyst

(AlBr_3) and pure bromine, and only gives mono-substitution. This demonstrates the activating effect of the $-\text{OH}$ group.

Nitration of phenol: Phenol reacts with dilute HNO_3 at room temperature (much milder conditions than benzene).

Exam Tip: In mechanism questions for electrophilic substitution, you must show: (1) the generation of the electrophile (e.g. NO_2^+ from $\text{HNO}_3/\text{H}_2\text{SO}_4$), (2) the curly arrow from the ring π system to the electrophile, (3) the Wheland intermediate with the $+$ charge on the ring, (4) loss of H^+ to restore aromaticity. For Friedel-Crafts, also show regeneration of the AlCl_3 catalyst.

3.4 Directing Effects

Substituents already on the benzene ring influence where the next electrophile attacks:

- **Electron-donating groups** ($-\text{OH}$, $-\text{NH}_2$, $-\text{CH}_3$, $-\text{OCH}_3$) are **2,4-directors** (ortho/para). They increase electron density at positions 2, 4, and 6, activating the ring.
- **Electron-withdrawing groups** ($-\text{NO}_2$, $-\text{COOH}$, $-\text{CHO}$) are **3-directors** (meta). They decrease electron density at positions 2, 4, and 6, so the electrophile attacks at position 3 (and 5).

4. Amines

4.1 Structure and Classification

Amine: An organic compound derived from ammonia (NH_3) by replacing one or more hydrogen atoms with alkyl or aryl groups. The nitrogen atom retains a lone pair of electrons, giving amines their characteristic basic and nucleophilic properties.

- **Primary amine (RNH_2):** One H replaced (e.g. methylamine CH_3NH_2 , phenylamine $\text{C}_6\text{H}_5\text{NH}_2$)
- **Secondary amine (R_2NH):** Two H replaced (e.g. dimethylamine $(\text{CH}_3)_2\text{NH}$)
- **Tertiary amine (R_3N):** Three H replaced (e.g. trimethylamine $(\text{CH}_3)_3\text{N}$)
- **Quaternary ammonium ion (R_4N^+):** All four H replaced, permanently charged (e.g. tetramethylammonium $(\text{CH}_3)_4\text{N}^+$)

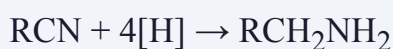
4.2 Preparation of Amines

Method 1 — From halogenoalkanes:



Conditions: Excess concentrated ammonia, heat in a sealed tube (to prevent NH_3 escaping). Excess ammonia minimises further substitution to secondary, tertiary amines, and quaternary ammonium salts.

Method 2 — Reduction of nitriles:



Conditions: LiAlH_4 in dry ether, then hydrolyse with dilute acid. Or H_2/Ni catalyst at high temperature and pressure. This produces a primary amine with one more carbon atom than the starting nitrile.

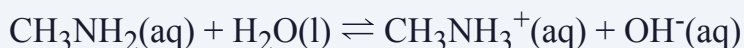
Method 3 — Reduction of nitrobenzene (for aromatic amines):



Conditions: Tin (Sn) + concentrated HCl, heat under reflux. Then add excess NaOH to liberate the free amine from its salt.

4.3 Basicity of Amines

Amines act as Bronsted-Lowry bases because the lone pair on nitrogen can accept a proton (H^+):



Basicity order: aliphatic amines > ammonia > phenylamine (aromatic amine)

- **Aliphatic amines** are stronger bases than ammonia because alkyl groups have a positive inductive effect (+I), pushing electron density towards the nitrogen. This makes the lone pair more electron-dense and more available to accept H^+ . The resulting cation (RNH_3^+) is also more stable due to the electron-donating alkyl groups.
- **Phenylamine** is a much weaker base than ammonia because the lone pair on nitrogen is partially delocalised into the benzene ring's π system. This delocalisation reduces the electron density on

nitrogen, making the lone pair much less available to accept H^+ .

Amines react with acids to form salts: $\text{CH}_3\text{NH}_2 + \text{HCl} \rightarrow \text{CH}_3\text{NH}_3^+\text{Cl}^-$ (methylammonium chloride)

4.4 Amines as Nucleophiles

Amines can act as nucleophiles because of their lone pair. They react with acyl chlorides and acid anhydrides to form amides (see section 2.3).

5. Polymers

5.1 Addition Polymers (Recap)

Formed from alkene monomers by opening the $\text{C}=\text{C}$ double bond. No atoms are lost — 100% atom economy. The repeating unit retains all atoms from the monomer. Examples: poly(ethene), poly(propene), PVC, PTFE. Non-biodegradable due to strong, non-polar $\text{C}-\text{C}$ backbone.

5.2 Condensation Polymers

Formed when bifunctional monomers (monomers with two functional groups) react together, with the loss of a small molecule (usually H_2O) at each link. Two main types:

Polyesters

Made from a dicarboxylic acid + a diol (or from a hydroxy acid with both $-\text{OH}$ and $-\text{COOH}$ on the same molecule):



The repeating linkage is the **ester bond** ($-\text{COO}-$).

Example: PET (Terylene) from benzene-1,4-dicarboxylic acid and ethane-1,2-diol. PET is used in drinks bottles, food packaging, and polyester clothing fibres.

Polyamides (Nylons)

Made from a dicarboxylic acid + a diamine:



The repeating linkage is the **amide/peptide bond (-CONH-)**.

Example: Nylon 6,6 from hexanedioic acid (6C) and 1,6-diaminohexane (6C). Used in clothing, carpets, ropes, and engineering plastics.

Natural polyamides: Proteins and silk are natural polyamides (polypeptides), with amino acids as the monomers.

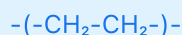
Addition vs Condensation Polymerisation

Addition Polymerisation

Monomer:



(alkene)



e.g. poly(ethene)

from ethene • C=C opens up • No by-product • Non-biodegradable • 100% atom economy • No polar bonds for hydrolysis

Condensation Polymerisation

Polyester: Diacid + Diol $\rightarrow$ $\text{-(OC-R-COO-R}'\text{-O)-}_n$ + $n\text{H}_2\text{O}$

ester linkage: -COO-

Polyamide: Diacid + Diamine $\rightarrow$ $\text{-(OC-R-CONH-R}'\text{-NH)-}_n$ + $n\text{H}_2\text{O}$

amide linkage: -CONH- • Small molecule lost (H_2O) • Can be hydrolysed (more biodegradable) • Two different monomers needed • Polar linkages broken by acid/base/enzymes

5.3 Biodegradability and Disposal

Property	Addition polymers	Condensation polymers
----------	-------------------	-----------------------

Biodegradability	Non-biodegradable (no polar bonds for enzymes to attack)	More biodegradable (ester/amide bonds can be hydrolysed by enzymes, acid, or base)
Hydrolysis	Cannot be hydrolysed	Hydrolysed by acids, bases, or enzymes
Environmental impact	Persists in landfill for centuries	Breaks down more readily
Disposal methods	Recycling, incineration, landfill	Biodegradation, hydrolysis, recycling

Exam Tip: To identify monomers from a condensation polymer: find the repeating unit, identify the linkage (ester $-\text{COO}-$ or amide $-\text{CONH}-$), break the linkage, and add back $-\text{OH}$ and $-\text{H}$ where the water was removed. Practice converting between polymer structure and monomer structures in both directions.

6. Amino Acids and Proteins

6.1 Structure of Amino Acids

Amino acids contain both an amine group ($-\text{NH}_2$) and a carboxylic acid group ($-\text{COOH}$) on the same carbon (the α -carbon). General formula: $\text{H}_2\text{N}-\text{CHR}-\text{COOH}$, where R is the variable side chain that determines the identity of each amino acid.

Zwitterion: A dipolar ion carrying both a positive and a negative charge, but overall electrically neutral. At the isoelectric point (the pH at which the net charge is zero), amino acids exist as zwitterions: $^+\text{H}_3\text{N}-\text{CHR}-\text{COO}^-$. The $-\text{NH}_2$ group is protonated (acting as a base) and the $-\text{COOH}$ group is deprotonated (acting as an acid).

Behaviour at different pH values:

- **Acidic solution (low pH):** The amino acid is fully protonated — $^+\text{H}_3\text{N}-\text{CHR}-\text{COOH}$. It carries a net positive charge (cation). It migrates towards the cathode (negative electrode) in electrophoresis.
- **Isoelectric point:** The zwitterion form $^+\text{H}_3\text{N}-\text{CHR}-\text{COO}^-$. No net charge; does not migrate in electrophoresis.

- **Basic solution (high pH):** The amino acid is fully deprotonated — $\text{H}_2\text{N}-\text{CHR}-\text{COO}^-$. It carries a net negative charge (anion). It migrates towards the anode (positive electrode).

All amino acids (except glycine, where $\text{R} = \text{H}$) have a chiral centre at the α -carbon (bonded to four different groups: $-\text{NH}_2$, $-\text{COOH}$, $-\text{H}$, and $-\text{R}$) and therefore exist as enantiomers. In nature, only L-amino acids are used to build proteins.

Zwitterions explain the high melting points of amino acids compared to similar-sized organic molecules: the ionic interactions between the $-\text{NH}_3^+$ and $-\text{COO}^-$ groups act like ionic bonds, requiring significant energy to break.

6.2 Peptide Bonds and Proteins

Amino acids link by condensation reactions, forming **peptide bonds ($-\text{CONH}-$)** with the loss of water:



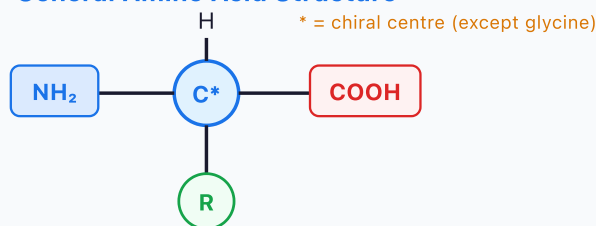
Two amino acids form a dipeptide. Three form a tripeptide. Many amino acids form a polypeptide (protein). The sequence of amino acids is the primary structure of the protein.

Hydrolysis of proteins/peptides: The peptide bond can be broken by:

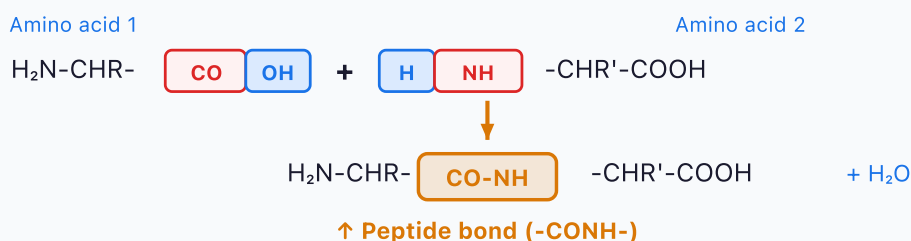
- **Acid hydrolysis:** Reflux with 6 mol dm^{-3} HCl for 24 hours. This breaks all peptide bonds, producing individual amino acids (as cations in acidic solution).
- **Enzymatic hydrolysis:** Specific enzymes (proteases/peptidases) catalyse hydrolysis at body temperature. Each enzyme is specific to particular amino acid sequences.

Amino Acid Structure and Peptide Bond Formation

General Amino Acid Structure



Condensation: Peptide Bond Formation



Exam Tip: You must be able to: (1) draw the zwitterion form of a given amino acid, (2) show the form in acid and base, (3) draw the dipeptide formed from two amino acids (showing the peptide bond and the water lost), (4) identify the number of possible dipeptides from two different amino acids (there are two: AA1-AA2 and AA2-AA1, because the peptide bond can form in either direction).

7. Organic Synthesis Routes

Organic synthesis involves converting one functional group into another using appropriate reagents and conditions. The key to synthesis questions is knowing the inter-conversions and being able to plan multi-step routes.

7.1 Key Functional Group Inter-conversions

Starting material	Reagent and conditions	Product
Alkene	HBr, room temperature	Halogenoalkane
Alkene	H ₂ O (steam), H ₃ PO ₄ catalyst, 300°C	Alcohol (industrial hydration)

Alkene	Br ₂ (in organic solvent)	Dibromoalkane
Halogenoalkane	Aqueous NaOH, reflux	Alcohol
Halogenoalkane	Ethanollic KCN, reflux	Nitrile (C chain +1)
Halogenoalkane	Ethanollic KOH, reflux (hot)	Alkene (elimination)
Halogenoalkane	Excess NH ₃ , heat, sealed tube	Primary amine
Primary alcohol	K ₂ Cr ₂ O ₇ /H ₂ SO ₄ , distil	Aldehyde
Primary alcohol	K ₂ Cr ₂ O ₇ /H ₂ SO ₄ , reflux	Carboxylic acid
Secondary alcohol	K ₂ Cr ₂ O ₇ /H ₂ SO ₄	Ketone
Alcohol	Conc. H ₂ SO ₄ or H ₃ PO ₄ , heat	Alkene (dehydration)
Aldehyde or ketone	NaBH ₄ in water/methanol	Alcohol (reduction)
Aldehyde or ketone	HCN + KCN catalyst	Hydroxynitrile (C chain +1)
Nitrile	LiAlH ₄ in dry ether (then H ₂ O)	Primary amine
Nitrile	Dilute acid or base, reflux	Carboxylic acid
Carboxylic acid + alcohol	Conc. H ₂ SO ₄ catalyst, reflux	Ester
Acyl chloride + alcohol	Room temperature	Ester (+ HCl)
Acyl chloride + NH ₃	Room temperature, excess NH ₃	Primary amide (+ NH ₄ Cl)
Benzene	Conc. HNO ₃ + conc. H ₂ SO ₄ , 50°C	Nitrobenzene
Nitrobenzene	Sn + conc. HCl, reflux, then NaOH	Phenylamine
Benzene	RCOCl + AlCl ₃ , reflux	Aryl ketone (Friedel-Crafts acylation)

Benzene	$\text{RCl} + \text{AlCl}_3$, reflux	Alkylbenzene (Friedel-Crafts alkylation)
---------	---------------------------------------	--

7.2 Strategy for Multi-Step Synthesis

1. **Work backwards (retrosynthetic analysis):** Start from the target molecule and ask "What could I make this from?" at each step.
2. **Identify the functional group changes needed** at each stage.
3. **Consider carbon chain changes:** If the target has more carbons than the starting material, you need a chain-extension step (nitrile formation via CN^- , or Friedel-Crafts, or Grignard reagent).
4. **Always state reagents AND conditions** for full marks.

Exam Tip: The most common synthesis routes examined involve: alkene $\rightarrow$ halogenoalkane $\rightarrow$ alcohol $\rightarrow$ aldehyde/ketone $\rightarrow$ carboxylic acid. Chain extension via halogenoalkane + $\text{KCN} \rightarrow$ nitrile $\rightarrow$ carboxylic acid (or amine) is frequently tested. You should be able to draw a complete synthesis map connecting all major functional groups.

8. NMR Spectroscopy

8.1 Carbon-13 (^{13}C) NMR

^{13}C NMR shows the number of different carbon environments in a molecule. Each chemically non-equivalent carbon produces a separate peak. The number of peaks tells you the number of distinct carbon environments.

Key features:

- Each peak represents one type of carbon environment (no splitting pattern information — peaks are singlets).
- The peak height does NOT indicate the number of carbons (unlike ^1H NMR integration).
- Symmetrical molecules have fewer peaks than carbon atoms (e.g. propanone CH_3COCH_3 has only 2 peaks: the two equivalent CH_3 carbons, and the $\text{C}=\text{O}$ carbon).

Carbon environment	Approximate δ / ppm
C-C (alkyl, R-CH ₃ , R-CH ₂ -R)	5-40
C-Cl, C-Br	25-55
C-O (alcohol, ether)	50-90
C=C (alkene)	110-140
C=C (aromatic)	120-160
C=O (aldehyde, ketone)	190-220
C=O (carboxylic acid, ester)	160-185

8.2 Proton (¹H) NMR

¹H NMR provides three key pieces of information about the hydrogen atoms in a molecule:

1. Chemical Shift (δ)

The position of the peak along the x-axis, measured in parts per million (ppm) relative to the reference standard TMS (tetramethylsilane, Si(CH₃)₄), which is defined as δ = 0.000 ppm.

Why TMS is used: (1) It gives a single, sharp peak. (2) Its peak is at a higher field than almost all organic protons, so it does not overlap. (3) It is chemically inert and does not react with samples. (4) It has a low boiling point (27°C) so it can be easily removed after analysis. (5) It dissolves in most organic solvents.

Proton environment	Approximate δ / ppm
R-CH ₃ (alkyl)	0.7-1.6
R-CH ₂ -R	1.2-1.8
R-CH ₂ -X (X = halogen)	3.0-4.2
R-O-CH ₃ or R-O-CH ₂ -	3.3-3.9

R-OH (alcohol)	0.5-6.0 (variable, often broad)
R-CHO (aldehyde)	9.0-10.0
Ar-H (aromatic/benzene ring)	6.5-8.0
R-COOH (carboxylic acid)	10.0-12.0
R-NH ₂ (amine)	1.0-4.5 (variable)

2. Integration (Relative Peak Areas)

The area under each peak is proportional to the number of hydrogen atoms in that environment. The ratio of peak areas gives the ratio of protons. For example, if the integration ratio is 3:2:1, the protons are in a 3:2:1 ratio (e.g. CH₃-CH₂-OH).

3. Splitting Pattern (Spin-Spin Coupling)

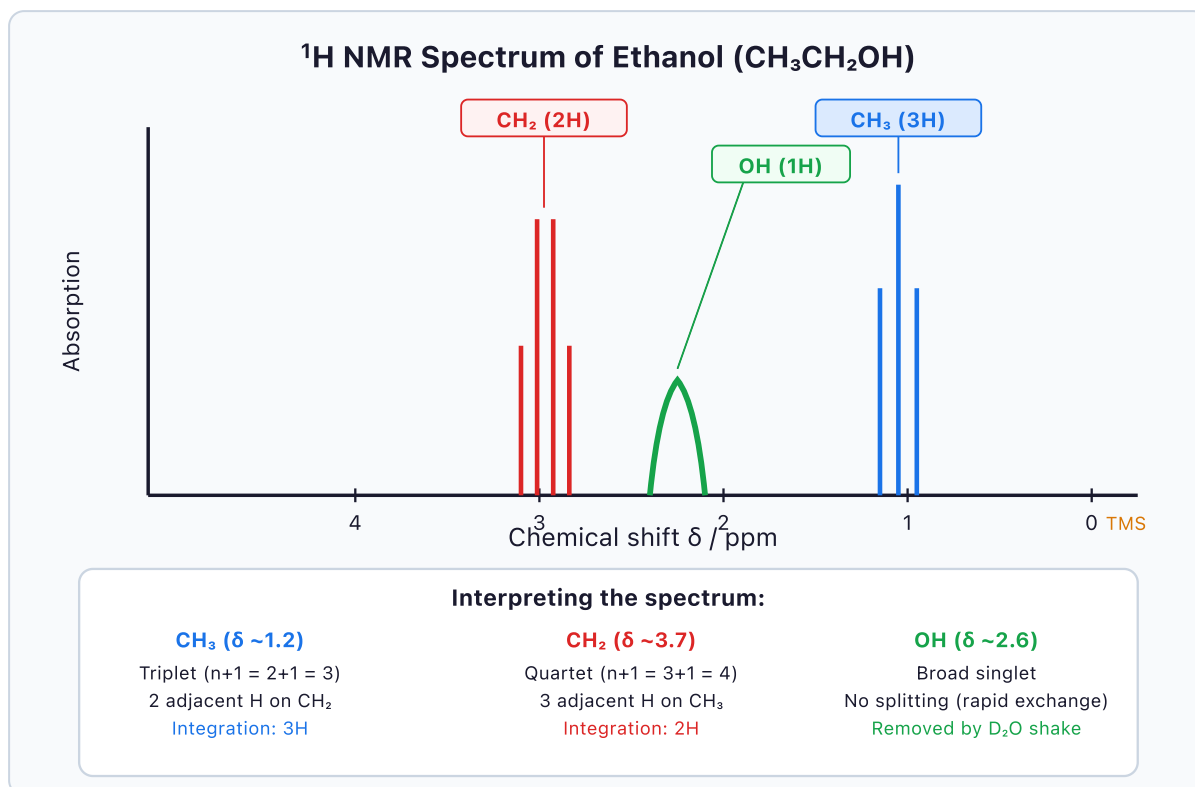
Protons on adjacent (neighbouring) non-equivalent carbons interact magnetically, causing peaks to split. The **n+1 rule** predicts the splitting: a proton with n equivalent neighbours on adjacent carbons is split into (n+1) peaks.

Number of adjacent H (n)	Splitting pattern (n+1)	Name	Relative peak heights
0	1 peak	Singlet (s)	1
1	2 peaks	Doublet (d)	1:1
2	3 peaks	Triplet (t)	1:2:1
3	4 peaks	Quartet (q)	1:3:3:1

Example — Ethanol (CH₃CH₂OH):

- CH₃ protons ($\delta \sim 1.2$): 3 equivalent H, split into a **triplet** by the 2 adjacent CH₂ protons. Integration = 3.
- CH₂ protons ($\delta \sim 3.7$): 2 equivalent H, split into a **quartet** by the 3 adjacent CH₃ protons. Integration = 2.

- OH proton ($\delta \sim 2.6$): 1H, appears as a broad **singlet**. It does not show splitting because of rapid proton exchange with water in the solvent.



8.3 D_2O Shake

Adding D_2O (deuterium oxide) to an NMR sample causes any O-H or N-H protons to exchange with deuterium. Since deuterium is not detected by ^1H NMR, the O-H or N-H peak **disappears** from the spectrum. This is a useful technique for identifying which peak corresponds to an exchangeable proton.

8.4 Using NMR to Determine Structure

Systematic approach:

- Determine the molecular formula (from mass spectrometry or elemental analysis).
- Calculate the degree of unsaturation ($\text{DBE} = (2\text{C} + 2 + \text{N} - \text{H} - \text{X}) / 2$). A DBE of 4 suggests a benzene ring.
- Count the number of peaks in ^{13}C NMR to find the number of carbon environments.
- In ^1H NMR: use chemical shift to identify proton types, integration for the number of each type, and splitting for the number of adjacent protons.
- Use D_2O shake to identify O-H or N-H protons.

6. Combine all evidence to propose a structure consistent with all data.

Exam Tip: (1) O-H and N-H protons show as broad singlets and are removed by D₂O shake. (2) The integration ratio directly gives the proton ratio. (3) Equivalent protons (by symmetry) give the same peak. (4) Chemical shift values are in the data booklet — you do not need to memorise them, but you must be able to use them. (5) In combined technique questions, use mass spec for molecular mass, IR for functional groups, and NMR for the detailed structure.

9. Chromatography

9.1 General Principles

Chromatography separates mixtures based on the different affinities of components for a **stationary phase** (which stays still) and a **mobile phase** (which moves). Components that interact more strongly with the stationary phase move more slowly; those that interact more with the mobile phase move faster. The degree of separation depends on the relative solubility/adsorption in each phase.

9.2 Thin Layer Chromatography (TLC)

- **Stationary phase:** Thin layer of silica (SiO₂) or alumina (Al₂O₃) coated on a glass or plastic plate (polar surface).
- **Mobile phase:** Organic solvent (e.g. ethanol, hexane, or mixtures).
- **Procedure:** A small spot of sample is placed on the baseline (drawn in pencil, not ink). The plate is placed in a beaker with a shallow layer of solvent. The solvent rises by capillary action, carrying components at different rates.

$$R_f = \text{distance moved by substance} / \text{distance moved by solvent front}$$

R_f values depend on the solvent, stationary phase, and temperature. They can be compared with known standards to identify substances. If spots are not visible, use a UV lamp (for UV-active compounds) or a locating agent (iodine vapour, ninhydrin for amino acids gives purple spots, potassium permanganate).

Uses: Quick identification of substances, monitoring the progress of a reaction, checking purity (pure substance = single spot).

9.3 Gas Chromatography (GC)

- **Stationary phase:** A high boiling-point liquid coated on an inert solid support inside a long coiled column (capillary column, up to 30 m).
- **Mobile phase:** An inert carrier gas (nitrogen, helium, or argon).
- **Procedure:** The sample is injected through a septum and vaporised in a heated injection port. The carrier gas carries the vaporised sample through the column. Components separate based on their boiling points and their affinity for the stationary phase. A detector records each component as it elutes from the column.

The chromatogram:

- **Retention time** (the time taken for a component to pass through the column) identifies each component by comparison with standards.
- **Peak area** is proportional to the amount (concentration) of that component — used for quantitative analysis.
- **Number of peaks** indicates the number of components in the mixture.

9.4 GC-MS (Gas Chromatography-Mass Spectrometry)

GC-MS combines gas chromatography (for separation) with mass spectrometry (for identification). After GC separates the mixture, each component enters the mass spectrometer, which produces a mass spectrum showing the molecular ion peak (M^+ , giving molecular mass) and fragmentation pattern (giving structural information). The mass spectrum is compared with a database for positive identification.

GC-MS is an extremely powerful analytical technique used in forensics (drug testing, arson investigation), environmental monitoring (pesticides, pollutants), food safety, and airport security.

9.5 Column Chromatography and HPLC

Column chromatography: Uses gravity to flow the mobile phase (liquid solvent) through a column packed with a solid stationary phase (silica/alumina). Used for preparative separation of larger quantities.

HPLC (High-Performance Liquid Chromatography): Similar to column chromatography but uses high pressure (~400 bar) to force the mobile phase through a column packed with very fine particles. This gives much faster and more efficient separation. A detector (usually UV) records the chromatogram.

Advantages of HPLC over GC:

- Can separate non-volatile compounds (GC requires compounds to be vaporised)
- Can separate thermally unstable compounds (GC uses high temperatures)
- Can analyse biological samples (proteins, drugs) in aqueous solution

Exam Tip: Know the key differences: TLC is qualitative, quick, and simple (identification by R_f). GC is quantitative (peak area) and separates volatile compounds (identification by retention time). GC-MS adds positive structural identification via mass spectrometry. HPLC is for non-volatile or heat-sensitive compounds. Always state the stationary and mobile phases when describing any chromatographic method.